ALIREZA KHEIRANDISH

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Google Scholar | GitHub | LinkedIn

Research Interests

Diffusion and generative models; inverse problems and computational imaging; uncertainty quantification; out-of-distribution detection; reinforcement learning; multimodal machine learning.

Education

Georgia Institute of Technology — Atlanta, GA
Ph.D. in Electrical and Computer Engineering — 2024–Present
Advisor: Prof. Sara Fridovich-Keil | GPA: 4.0/4.0

Georgia Institute of Technology — Atlanta, GA
M.S. in Electrical and Computer Engineering — 2026
Thesis: *KLIP: Localized Distribution Shift Detection via KL-Divergence with Diffusion Priors in Inverse Problems*
GPA: 4.0/4.0

Sharif University of Technology — Tehran, Iran
B.S. in Electrical Engineering (Communications) — 2018–2023
GPA: 18.99/20.00 | Ranked 6th among 176 electrical engineering students.

Peer-Reviewed Publications

KLIP: Localized Distribution Shift Detection via KL-Divergence with Diffusion Priors in Inverse Problems
Alireza Kheirandish, Jihoon Hong, Sara Fridovich-Keil.
CVPR 2026 | *Paper* | *Code*

LLM-Augmented Symbolic RL with Landmark-Based Task Decomposition
Alireza Kheirandish, Duo Xu, Faramarz Fekri.
IEEE ICASSP 2025 | *Paper* | *Code*

GT-NLP at SemEval-2025 Task 11: EmoRationale, Evidence-Based Emotion Detection via Retrieval-Augmented Generation
Daniel Saeedi, **Alireza Kheirandish**, Sirwe Saeedi, Hossein Sahour, Aliakbar Panahi, Iman Naeeni.
SemEval 2025 *(ACL Anthology)* | *Paper* | *Code*

Preprints

LACToS: Local Anomaly Detection via Conditional Typicality of Superpixels
Jihoon Hong, **Alireza Kheirandish**, Rohan Sanda, Sara Fridovich-Keil.
arXiv:2610.31538*, 2026* | *Preprint* | *Code*

SORD: Support-guided OOD Removal with Diffusion
Alireza Kheirandish, Sara Fridovich-Keil.
arXiv:2610.33194*, 2026* | *Preprint* | *Code*

Research Experience

VOILA Lab, Georgia Institute of Technology — Atlanta, GA
Research in diffusion models, inverse problems, and localized distribution shifts

KLIP: Localized Distribution Shift Detection (CVPR 2026)
- Introduced a calibration-free spatial KL-divergence metric between clean diffusion priors and measurement-conditioned posteriors to localize distribution shifts directly from inverse-problem measurements.

SORD: Support-Guided OOD Removal with Diffusion (arXiv 2026)
- Developed support-guided diffusion restoration that exploits available measurements to remove localized structured corruption in images and audio without paired training data or a learned corruption model.

LACToS: Local Anomaly Detection via Conditional Typicality of Superpixels (arXiv 2026)
- Co-developed a superpixel-level conditional typicality detector using an efficient statistical proxy; evaluated robustness across medical imaging, faces, and X-ray security screening.

Research Experience (continued)

SENTINEL Research Lab, Georgia Institute of Technology Atlanta, GA
Symbolic reinforcement learning and large language models

- Developed an LLM-augmented symbolic reinforcement learning framework for decomposing complex tasks into subtasks and generating first-order logic rule templates.
- Combined trajectory-based subtask identification with inductive logic programming to refine symbolic policies and reduce reliance on hand-specified rules.

Multilingual Emotion Detection Research Georgia Tech
Retrieval-augmented generation for NLP

- Co-developed EmoRationale, which retrieves semantically relevant examples for LLM-based multilingual emotion classification and evidence-based predictions without task-specific fine-tuning.

Digital Signal Processing Research Lab, Sharif University of Technology Tehran, Iran
Undergraduate research in graph signal processing

- Developed two graph-signal denoising methods using Graph Empirical Mode Decomposition (GEMD), with an emphasis on low signal-to-noise-ratio settings.

Selected Projects

Multimodal Protein Mutation Effect Prediction GitHub
Developed a model combining protein sequence and structural representations using graph-based and attention-based learning to predict mutation effects.

Wearable EEG/EOG Drowsiness Detection Glasses GitHub
Designed and fabricated a 3D-printed wearable prototype with embedded circuitry for EEG/EOG sensing and drowsiness monitoring.

Teaching Experience

Teaching Assistant, Georgia Institute of Technology
Foundations of Computational Imaging (Fall 2025); Circuit Analysis / ECE 2040 (2024).

Additional Teaching Assistantships
Probabilistic Graphical Models in Machine Learning; Probability and Statistics; Artificial Intelligence and Biological Computation; EEG Signal Processing.

Technical Skills

Programming: Python, C/C++, MATLAB | **Machine Learning:** PyTorch, TensorFlow
Research: Diffusion models, inverse problems, signal processing, computational imaging, reinforcement learning
Tools: Git, Linux, LaTeX

Honors & Awards

- Ranked 159th among approximately 144,000 students in Iran's National University Entrance Examination (Mathematics/Physics), 2018.
- Ranked 6th among 176 electrical engineering students at Sharif University of Technology (2018 cohort).

Leadership & Service

Iranian Student Association, Georgia Tech: Board Member and Community Outreach Chief.
Sharif University of Technology Association (SUTA), Atlanta: Chapter Board Member and Treasurer.

Languages

Persian (native); English (professional proficiency).